Selective Modernization and Bounded Responsiveness in Japan's Waste-Incineration Fleet: A Facility-Level Panel Study

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Abstract

Japan relies heavily on municipal waste incineration, yet a large share of its incineration fleet still does not convert waste heat into electricity. A one-average-fleet view therefore risks conflating two different questions: who records observed transition into generation at all, and how plants perform once they already generate. Prior work often studies fleets in aggregate or studies generators only, which blurs those two estimands. This paper separates them in a national facility-level panel built from Ministry of the Environment data for FY2005–FY2024. It first models observed transition into generation among coded facilities first seen without it, then models energy recovery efficiency among operating generators. Transition is selective rather than diffuse: in the main lagged hazard model, facilities older than 10 years are about 1–2 percentage points less likely than 0–10 year facilities to record transition in the next observed year, while each additional 100 t/day of design capacity raises annual transition probability by about 0.5 percentage points. Within the generating segment, efficiency is lower at older plants and higher at larger, more fully utilized ones, and the within-to-total variance ratio of log-efficiency is about 0.15. Together, the results show that observed transition into generation is selective, whereas performance within generation is bounded by strong cross-facility heterogeneity. For comparable municipal fleet studies, the broader lesson is methodological: adoption and conditional performance should not be treated as one average process.

Keywords: waste incineration; waste-to-energy; Japan; energy recovery; facility panel; transition

1 Introduction

Japan operates one of the world's most incineration-dependent municipal waste systems, yet a large share of the fleet still burns waste without generating electricity from the heat it produces (Ministry of the Environment Japan, 2022; Tabata & Tsai, 2016). The transition problem is therefore not whether incineration exists, but which facilities move into useful energy recovery and what performance looks like once they do.

That pattern is easy to flatten into one average fleet story. But the relevant questions are not the same. One is extensive: which facilities record observed transition into power generation at all? The other is intensive: among plants that already generate, which facilities achieve high energy recovery efficiency? A single fleet-average view obscures the difference between entering generation and performing within generation.

This paper separates transition into generation from conditional performance within the generating segment in a national facility-level panel. Its originality is not simply that it studies Japan, but that

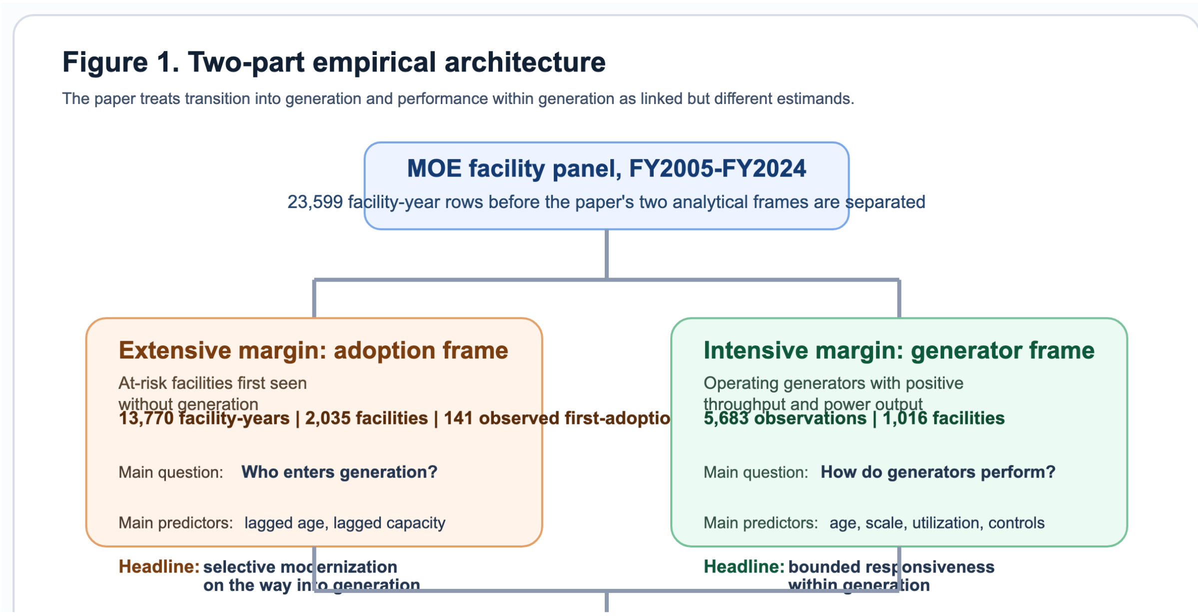

Figure 1: Two-part empirical architecture linking the adoption frame, generator frame, and the paper's synthesis claim.

it uses one facility-level dataset to answer a question that adjacent literatures usually split apart: who records observed entry into generation, and what constrains performance once generation already exists. On the adoption margin, observed transition is selective rather than diffuse: younger and larger facilities are more likely to record transition, while old and small plants rarely do. On the intensive margin, efficiency among generators is strongly structured by age, scale, and utilization, and within-facility movement is small relative to between-facility heterogeneity. Read together, those results imply a split transition rather than one average modernization process.

The gap addressed here is therefore narrower than a generic claim that Japan has been understudied. Fleet-level waste-to-energy studies can describe system trajectories, while generator-only studies can explain conditional performance. What they do not show within one municipal fleet is whether entry into generation and performance within generation follow the same modernization logic. Japan makes that problem visible because the fleet remains large, municipally embedded, and technologically heterogeneous, with a substantial non-generating segment still active alongside a smaller modern generating segment. The broader claim remains narrow: in similar fleet-transition settings, separate models of adoption and conditional performance may reveal patterns that a single average-fleet view misses.

The rest of the paper proceeds as follows. Section 2 positions the paper against the literature most relevant to the analytical split. Section 3 introduces the two linked analytical frames and the main estimation choices. Section 4 reports the adoption and efficiency results in sequence, then ties them together. Section 5 interprets the combined finding, explains what the data still cannot identify, and states a short set of evidence-consistent implications. Section 6 concludes.

Figure 1 makes the paper's core move explicit. The adoption layer uses the coded at-risk frame to model observed entry into generation, while the efficiency layer uses the canonical generator frame to model conditional performance once generation already exists. The point is not to stack two separate studies in one article, but to show that a one-average-fleet view would otherwise force entry and performance into the same modernization story.

2 Literature Positioning

This paper speaks to three literatures: waste-to-energy systems, facility-level efficiency analysis, and infrastructure lock-in. The point of the review is not to survey them exhaustively, but to show why the present question requires an explicit split between adoption into generation and conditional performance within generation.

Work on waste-to-energy systems often documents national trajectories, technology choices, or lifecycle implications of thermal treatment (Astrup et al., 2015; Sun et al., 2018). That work is important for sectoral context, but it often treats the incineration fleet as a category. It can show whether energy recovery is growing, yet it is less well suited to distinguishing facilities that enter generation from those that do not.

Facility-level efficiency studies are closer to the present paper, but they typically begin with plants that already generate or focus on technical performance conditional on operation. Recent Chinese plant-level work, for example, is highly informative about performance differentials within the generating segment (Cui et al., 2026), but it does not directly address the modernization margin between non-generating and generating facilities. In practical terms, such studies answer the intensive-margin question while leaving the extensive-margin question open.

The lock-in literature adds a different expectation: infrastructure performance may be shaped by durable design choices, inherited scale, and institutional arrangements rather than by frequent large reversals at mature facilities (Unruh, 2000; Seto et al., 2016). The useful implication here is empirical rather than grand-theoretical. If design-conditioned heterogeneity dominates, cross-facility differences should matter more than large within-facility reversals over time. That expectation is only partially visible if entry into generation and performance within generation are never separated.

The paper therefore contributes by bringing those strands together in one facility-level design. It asks first who records observed transition into generation and then asks how the generating segment performs once generation already exists. In practical terms, it does what adjacent fleet-average and generator-only studies do not: it models observed entry and conditional performance within the same municipal fleet. Separating those margins, rather than identifying any single mechanism, is the paper's main originality claim.

3 Data and Design

The analysis uses the Ministry of the Environment's General Waste Treatment Survey for FY2005–FY2024 (Ministry of the Environment Japan, 2022). The full panel contains 23,599 facility-year rows. Within that, the coded full-fleet frame contains 19,827 observations across 2,948 facilities with usable identifiers. This paper uses two linked samples because one sample cannot answer both parts of the transition problem.

The first analytical frame is the coded adoption frame. It includes facilities first observed without

Table 1: Analytical frames used in the integrated paper design

Analytical frame	Observations	Facilities	Events	Role in paper
Coded full-fleet frame	19,827	2,948	–	Starting point after facility identifiers are resolved
Adoption risk set	13,770	2,035	141	Facilities first observed without power generation
Lagged adoption model frame	11,717	1,915	140	Main hazard model with lagged age and capacity predictors
Canonical generator frame	5,683	1,016	–	Operating generators used for conditional efficiency models

Note: the adoption and generator frames are linked but not interchangeable; each sample is defined to answer a different margin of the transition problem.

power generation and follows them until they either record observed transition into generation or remain non-generating in the panel window. After excluding left-censored facilities already generating in their first observed year, the adoption risk set contains 13,770 facility-years across 2,035 facilities, with 141 observed first-adoption events. The main adoption model is a lagged discrete-time logit hazard estimated on 11,717 observations across 1,915 facilities and 140 retained events. Predictors are prior-year age band and prior-year design capacity, with year fixed effects, prefecture fixed effects, and facility-clustered standard errors. This is an observed-transition model, not a complete structural model of all possible modernization pathways.

The second analytical frame is the canonical generator frame. It contains operating facilities with positive throughput and positive power output, after standard cleaning and a bounded efficiency measure. The regression frame contains 5,683 observations across 1,016 facilities. The dependent variable is winsorized log electricity generated per tonne processed. Main predictors are facility age, design capacity, capacity utilization, waste heating value, and a grid-emission control. The main specifications are pooled OLS, year fixed effects, random effects, and year fixed effects plus random effects, all used as structured descriptive models rather than clean structural estimates (Wooldridge, 2010).

The two layers belong in one paper because they answer sequential parts of the same modernization problem. The adoption layer identifies which facilities appear to enter the generating regime at all. The efficiency layer identifies whether large gains remain available once facilities are already inside that regime. Without the first layer, the paper would reduce transition to generator performance alone. Without the second, it would say who enters generation but not whether large efficiency gaps remain inside the generating segment.

The main identification limits are explicit. In the adoption layer, the paper models observed transition within the coded risk set, not unrestricted fleet-wide modernization. In the efficiency layer, age is closely tied to time and within-facility movement is limited, so the defended interpretation is one of structured conditional association rather than strict causal identification. The paper therefore aims for disciplined inference rather than maximal causal reach. All reported effects should be read as conditional associations within the specified samples, not as estimates of structural causality or policy effects. The paper does not claim that the low within-facility variance ratio resolves all fixed-effects concerns; instead, it uses that variance structure to motivate why a cross-facility descriptive model remains substantively useful for the question at hand.

Table 2: Main lagged hazard results for observed transition into generation

Variable	AME (pp)	SE (pp)
Prior-year age 10–20 yrs (vs 0–10)	-1.76	0.28
Prior-year age 20–30 yrs (vs 0–10)	-1.72	0.42
Prior-year age 30+ yrs (vs 0–10)	-1.13	0.39
Prior-year capacity (per 100 t/day)	0.50	0.20
Model summary		
Observations		11,717
Facilities		1,915
First-adoption events		140
Pseudo-R-squared		0.1842

Note: entries are average marginal effects in percentage points from the main lagged logit hazard with year and prefecture fixed effects and facility-clustered standard errors.

4 Results

4.1 Adoption into generation is selective rather than diffuse

The adoption results show a strongly selective transition pattern. In the risk set, annual event rates collapse after age 10 and rise sharply across capacity quartiles. Facilities aged 0–10 years account for 102 first-adoption events, while the three older age bands together account for only 39. By capacity, the largest quartile accounts for 99 first-adoption events, whereas the smallest quartile records only 1.

The discrete-time logit hazard compresses that pattern into a clear main result. Relative to 0–10 year facilities, plants aged 10–20 years are about 1.76 percentage points less likely to record transition in the next observed year, plants aged 20–30 years are about 1.72 percentage points less likely, and plants aged 30 years or more are about 1.13 percentage points less likely. Each additional 100 t/day of prior-year design capacity raises annual transition probability by about 0.50 percentage points. The sign pattern is stable in the complementary log-log and linear probability robustness variants.

This is not the pattern one would expect from diffuse late-life conversion of the old small-plant segment. Observed transition is concentrated among facilities that were already younger and larger before the event year. The extensive margin therefore looks like selective modernization rather than broad catch-up across the whole fleet.

The pathway audit supports that interpretation without turning it into a stronger mechanism claim than the data earn. Among the 141 observed adoption events, 82 are classified as reset- or rebuild-like, 38 as continuity-type upgrades, 20 as forward-dated or placeholder entries, and 1 as unresolved. This is descriptive pathway evidence, not mechanism identification. That distribution shows that capital-side selectivity is empirically present, but it does not uniquely identify replacement, major refurbishment, or new build as the single pathway. In the main text, the audit therefore functions as a credibility guard rather than as a coequal source of originality.

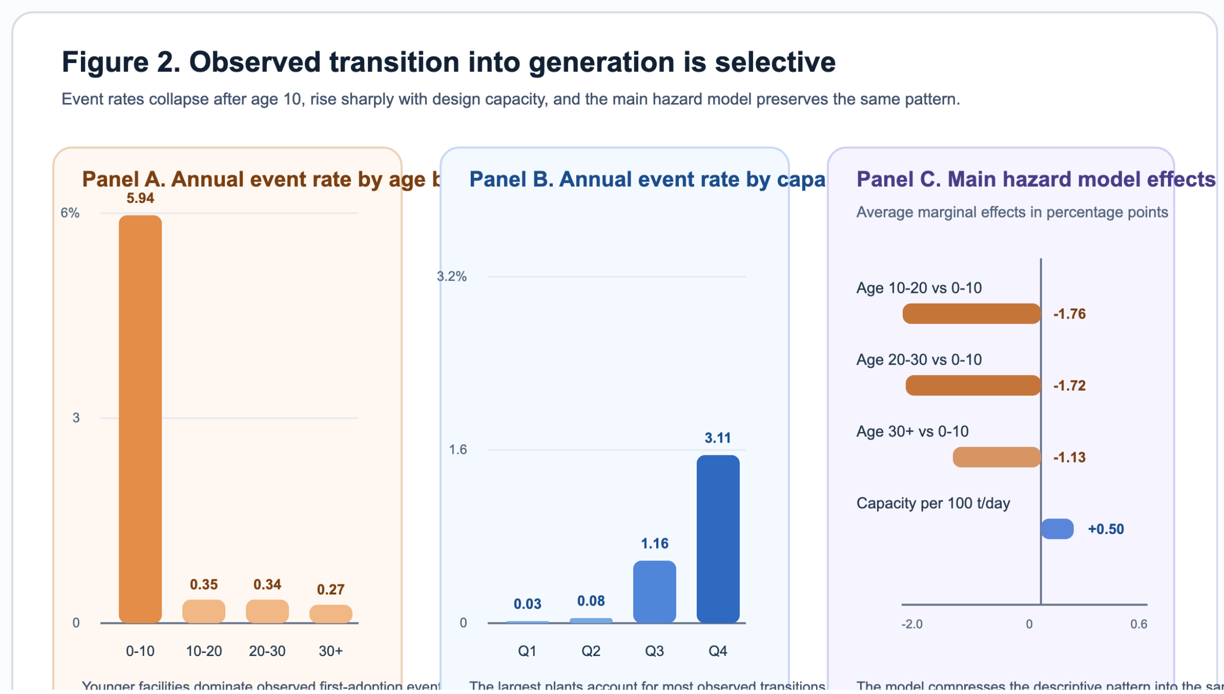

Figure 2: Adoption event rates by age band and capacity quartile, paired with the main hazard-model effects.

4.2 Performance within generation is bounded and strongly structured

The generator results tell a different but complementary story. In the canonical generator frame, efficiency is consistently lower at older facilities and higher at larger and more fully utilized ones. Across the main specifications, the age coefficient remains negative, the capacity coefficient remains positive, and the utilization coefficient remains positive. The magnitudes differ across models, but the sign pattern is stable, and the emphasis stays on structured association rather than on any single structural parameter.

The strongest descriptive result is not one coefficient but the variance structure. The within-to-total variance ratio of pooled log-efficiency is 0.1499. In other words, the large majority of variation in the dependent variable is between facilities rather than within facilities over time. The ratio remains low in both the pre-Fukushima and post-Fukushima windows, falling from 0.1795 before 2011 to 0.0956 after 2011. That pattern does not prove irreversibility, but it is hard to reconcile with a world in which mature generators frequently undergo large late-life reversals that reshape the fleet distribution.

The efficiency margin therefore looks bounded rather than static. Facilities do respond within a design envelope, especially through utilization and operational discipline, but the cross-sectional hierarchy remains strong. Older facilities systematically underperform younger ones; larger plants systematically outperform smaller ones; and utilization matters more as a supporting lever within the generating segment than as a fleet-wide equalizer.

This is why the intensive margin cannot be inferred from the adoption margin alone. A facility can

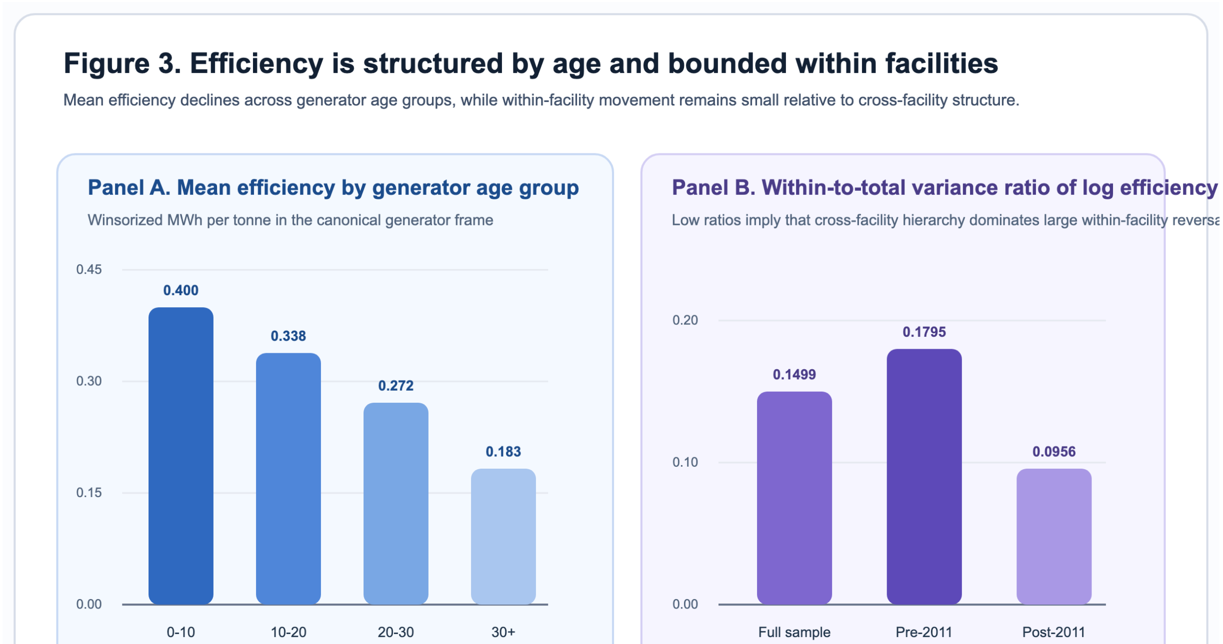

Figure 3: Mean efficiency declines across generator age groups, while the within-to-total variance ratio stays low in the full sample and in pre/post-Fukushima splits.

be inside generation without being close to the frontier, yet the evidence also suggests that operations alone are unlikely to erase large vintage and scale gaps once plants are mature. Conditional performance is therefore a bounded-performance problem rather than a simple continuation of the entry problem.

4.3 Why the two results belong together

Read together, the two margins change the story the fleet appears to tell. The adoption results show that entry into generation is already selective before conditional efficiency is considered, while the efficiency results show that large performance gaps inside the generating segment are not easily erased through within-facility movement alone. A single fleet-average model would ask one modernization narrative to explain both who enters generation and how mature generators perform. The two-part design instead shows selective modernization on the way into generation and bounded responsiveness once generation already exists.

5 Discussion

The paper's first interpretive claim is methodological. Japan's incineration transition should not be modeled as one average process because the full-fleet transition question and the generator-performance question are linked but not the same estimand. The adoption layer identifies who appears to move into generation. The efficiency layer identifies how the generating segment performs once it exists. That distinction is the paper's main contribution.

Table 3: Core conditional-efficiency specifications in the canonical generator frame

Variable	Model 1 Pooled OLS	Model 2 Year FE	Model 3 RE	Model 4 Year FE + RE
Facility age (years)	-0.0279***	-0.0348***	-0.0188***	-0.0332***
	(0.0022)	(0.0022)	(0.0025)	(0.0021)
Capacity (100 t/day)	0.0874***	0.1030***	0.0405***	0.0519***
	(0.0083)	(0.0086)	(0.0083)	(0.0096)
Capacity utilization	0.7468***	0.7789***	0.6199***	0.5411***
	(0.1421)	(0.1346)	(0.0997)	(0.0943)
Heating value (MJ/kg)	0.0010	0.0032	0.0006	0.0012
	(0.0023)	(0.0021)	(0.0012)	(0.0010)
Grid EF (kg-CO_2/kWh)	0.3182	-0.4466	1.6333***	-0.1951
	(0.2219)	(0.2714)	(0.1965)	(0.2101)
Observations	5,683	5,683	5,683	5,683
Facilities	1,016	1,016	1,016	1,016
R-squared	0.2470	0.3721	0.1647	0.3076

Note: standard errors are in parentheses. *** $p < 0.01$. Coefficients are reported as structured conditional associations rather than as strict structural parameters.

The substantive interpretation is correspondingly two-part. On the adoption margin, the data do not support a story of broad late conversion among old small plants. Observed transition is concentrated among younger and larger facilities, and the pathway audit shows more reset- or rebuild-like cases than continuity-type upgrades. On the efficiency margin, age, scale, and utilization matter strongly, while within-facility movement remains modest relative to the cross-sectional hierarchy. The evidence is therefore most consistent with selective capital-side modernization at the weak end of the fleet and bounded responsiveness within the generating segment.

That interpretation should remain calibrated. The pathway audit does not prove that replacement is the unique pathway of modernization. The regression results do not provide strict causal estimates of vintage lock-in. Alternative interpretations remain possible, including reporting compression, unobserved retrofit histories, and institutional constraints that limit operational responses. The defended claim is narrower: the data support a selective modernization process and bounded performance envelope, not a uniquely identified mechanism or a full causal hierarchy.

These are implications, not estimated policy rankings. For the weakest segment, especially older non-generators and small plants, the evidence points more toward capital-renewal planning than toward diffuse late-life operational improvement. For the already-generating segment, utilization, routing, and selective upgrading remain real levers, but they appear more likely to preserve or modestly improve performance within the existing envelope than to eliminate large inherited gaps.

6 Conclusion

Japan's incineration transition is not one smooth modernization process. It is split between selective entry into generation and bounded performance among facilities already inside the generating regime. Within the coded adoption frame, observed transition into generation is selective rather than diffuse. Within the canonical generator frame, efficiency is bounded by age, scale, and utilization, with cross-facility heterogeneity dominating large within-facility reversals.

The paper's main contribution is to show that these are distinct but linked questions and that treating them as one average fleet process obscures both. The evidence does not identify one unique modernization pathway or one optimal intervention ranking. It does indicate that the weakest segment of the fleet looks more like a capital-side modernization problem, whereas the generating segment looks more like a bounded-performance problem. For other municipal fleet-transition studies, that analytical separation may be more informative than another fleet average.

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Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey. Processed study outputs, manuscript figures, and the associated reproducible analysis workspace can be provided by the author on reasonable request.

Generative AI and AI-Assisted Technologies Statement

During the preparation of this manuscript, the author used OpenAI Codex and Anthropic Claude to support drafting, language revision, and organizational planning. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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